EMI Bounce Likelihood Model

Internship Project Report, Canara Bank

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Note: the bank could not share its data, so this project uses a synthetic dataset built to match the bank's schema. All results show that the method works. They are not measurements of real customers.

1. Objective

The objective of the Bounce Likelihood Model is to predict the probability of a near-term EMI bounce or missed payment for each borrower, so that the bank can act before the due date.

A bounce happens when the account balance is too low on the EMI due date. Bounces create penalty and follow-up costs and push customers towards the SMA and NPA stages. A model that ranks customers by risk lets the collections team send reminders to the right people at the right time.

Problem type: binary classification (TARGET_BOUNCE = 1 if the EMI bounced, 0 if paid on time), with the model producing a risk score and a probability for each account.

2. Data

The dataset has **200,000 loan accounts** (200,400 rows before duplicates were removed) and **16 columns**. Only standard (regular) accounts are in scope, so the dataset has no asset-class column.

Column	Description
V_ACCOUNT_NUMBER	Loan account number (identifier)
V_CUST_REF_CODE	Customer ID (some customers hold more than one loan)
SANCTIONED_LIMIT	Loan amount sanctioned (INR)
CURRENT_OUTSTANDING	Principal still to be repaid

Column	Description
EMI_AMT	Monthly instalment
D_FIRST_DISBURSEMENT_DATE	Date of first disbursement
LOAN_AGE_MONTHS	Tenure of the loan, in months
UTILIZATION_PERCENTAGE	Outstanding as a percentage of the sanctioned limit
LIMIT_AVAILABLE	Sanctioned limit minus outstanding
AVG_BAL_PREV_MNTH / PREV_QTR / FY	Average account balance for the previous month, quarter and financial year
CNT_MOV_1M / 3M / 6M	Count of credit (CRR) movements over 1, 3 and 6 months
TARGET_BOUNCE	Target: 1 = EMI bounced, 0 = paid

Target balance: about 7.03% of accounts bounced (14,085 bounces and 186,315 non-bounces in the 200,400 rows). The data is **imbalanced**, so accuracy is not a suitable measure.

Synthetic data. The target was generated from a hidden risk rule plus random noise. Risk rises when the balance is low relative to the EMI, credit activity is falling, utilization is high, or the loan is new. The data also includes realistic problems on purpose: missing values, duplicate rows, overdrawn accounts and extreme outliers.

3. Data Cleaning

Issue	Finding	Action
Data types	Account number read as a number, date read as text	Account number converted to text, date converted to a date
Duplicates	400 duplicate rows	Removed, leaving 200,000 unique accounts

Issue	Finding	Action
Missing values	AVG_BAL_PREV_MNTH 2,999; AVG_BAL_PREV_QTR 4,017; AVG_BAL_FY 5,832; CNT_MOV_6M 2,031; UTILIZATION_PERCENTAGE under 1%	Utilization rebuilt from outstanding and sanctioned limit. Other columns flagged and filled with the median inside the modelling pipeline, after the train/test split
Logic checks	Outstanding not above sanctioned, utilization within 0 to 100, movement counts in order (1M, 3M, 6M)	All checks passed
Negative balances	1,200 accounts (0.6%)	Kept. These are overdrawn accounts and carry real risk
Outliers	A few AVG_BAL_FY values of up to INR 8.3 crore, about 15 times the 99.9th percentile	Cross-checked against the quarterly balance. Rows with an FY balance more than 10 times the quarterly balance were treated as errors and set to missing. Genuine high-balance customers, whose three balances agreed, were kept

Why fill after the split: computing medians on the full data would let information from the test set leak into training. Filling inside the pipeline avoids this.

4. Exploratory Data Analysis

Target. About 93% of accounts paid on time and 7% bounced, so the data is strongly imbalanced.

Distributions. The money columns (sanctioned limit, outstanding, EMI, balances) are strongly right-skewed. For example, the typical monthly balance is about INR 52,000 (median), against a mean of about INR 1.41 lakh. Log scales were used for these charts. The three balance columns look almost identical, and so do the three movement columns, so they carry overlapping information.

Bouncers compared with non-bouncers (medians):

Feature	Paid	Bounced	Reading
AVG_BAL_PREV_MNTH	INR 55,217	INR 20,625	Bouncers hold about 63% less balance
UTILIZATION_PERCENTAGE	63.5%	72.6%	Bouncers still owe more of the loan
CNT_MOV_1M	15	11	Bouncers receive fewer credits
LOAN_AGE_MONTHS	25	20	Bouncers have younger loans

Correlation with the target (Spearman). The raw features are individually weak, which is normal for a rare behavioural target: AVG_BAL_PREV_MNTH about -0.17, CNT_MOV_1M about -0.11, UTILIZATION_PERCENTAGE about +0.08. A negative sign means more of the feature goes with a lower bounce risk. Decile charts showed that these weak correlations hide strong patterns: the bounce rate falls from about 18% in the lowest balance decile to about 1.6% in the highest.

Key EDA findings

1. Only 7% of accounts bounce, so the data is imbalanced.
2. Money columns are skewed, so medians and log transforms are appropriate.
3. Bouncers have lower balances, fewer credits and slightly higher utilization.
4. No single raw feature separates the groups, so ratio features and a proper model are needed.

5. Feature Engineering

Feature	Formula	Business question
BAL_TO_EMI	Monthly balance divided by EMI	Can the customer afford the EMI now?
EMI_TO_SANCTION	EMI divided by sanctioned limit	How heavy is the instalment?
BAL_TREND	Monthly balance divided by quarterly balance	Is the balance draining (below 1) or growing?
MOV_MOMENTUM	(1M count x 3) divided by 3M count	Is credit activity slowing (below 1)?

Feature	Formula	Business question
EARLY_LOAN	1 if loan age is under 6 months	Is the loan in its early, riskier months?

Divisions were guarded against zero, and infinite values were converted to missing.

Result. BAL_TO_EMI is the strongest single signal. The bounce rate across its ten equal-sized groups falls steadily:

Balance covers (EMIs)	Bounce rate
Under 1 (or overdrawn)	25.3%
1 to 1.5	13.4%
2.1 to 2.7	6.8%
4.4 to 5.7	3.0%
7.8 to 11.9	1.2%
Above 11.9	0.6%

The lowest group bounces about 40 times more often than the highest. Its correlation with the target (-0.25) is stronger than that of any raw column.

6. Modelling Approach

Split. 80% train (159,999 accounts) and 20% test (40,001 accounts). The split was **stratified** (both sets have a 7.03% bounce rate) and **grouped by customer**, so no customer appears in both sets (overlap checked: 0).

Metrics. Accuracy was not used, because a model that predicts "no bounce" for everyone would score 93%. The metrics were:

- **ROC-AUC:** how often a random bouncer is scored above a random non-bouncer (0.50 is a useless model).
- **PR-AUC:** how well the model finds bouncers without too many false alarms (a useless model scores about 0.07).
- **Top-20% capture:** the share of all bouncers reached by contacting the 20% highest-scored customers (random contact reaches 20%).

Imbalance handling. Class weights (Logistic Regression, Random Forest) and scale_pos_weight (XGBoost, LightGBM) so that each bouncer counts about 13 times more during training.

Models compared. Logistic Regression (with signed-log transform and scaling), Random Forest, Gradient Boosting, XGBoost and LightGBM.

7. Results

Cross-validation (customer-grouped folds on the training data) gave a ROC-AUC of about 0.81 and a PR-AUC of about 0.29 to 0.30 for **all five models**, with differences smaller than the fold-to-fold variation. The models are statistically tied.

Hyperparameter tuning. A randomised search for LightGBM (15 combinations) did not improve on the original settings. The test scores were nearly identical:

Model (full training data, test set)	ROC-AUC	PR-AUC	Top 20% captures
LightGBM (original settings)	0.810	0.293	61.6%
LightGBM (tuned)	0.810	0.291	61.2%
Logistic Regression	0.809	0.286	61.2%

Final model: LightGBM with the original settings (300 trees, learning rate 0.05, 15 leaves, at least 100 customers per leaf, 80% row and feature sampling). It was chosen because it is marginally best, handles missing values natively and scores quickly. Logistic Regression is kept as the simple, explainable benchmark.

Interpretation. A ROC-AUC of 0.81 means the model scores a bouncer above a non-bouncer about 81% of the time. A PR-AUC of 0.29 is about four times better than the 0.07 of a random model. By contacting the top 20% of customers, the bank reaches about 62% of bouncers, which is 3 times the efficiency of random contact.

Calibration. Because bouncers were up-weighted during training, the raw scores overstated risk: the average predicted bounce rate was 35.8%, against an actual rate of 7.0%. An isotonic calibrator fitted on a held-out part of the training data brought the average predicted rate to 7.0%. Calibration does not change the ranking, so recall and lift are unchanged.

8. Threshold and Business Use

The threshold is the score above which an account is flagged. Flagging more customers catches more bouncers but wastes more calls.

Flag top	Accounts flagged	Bouncers caught	Recall	Precision	Lift
5%	2,001	744	26.4%	37.2%	5.3x
10%	4,001	1,170	41.6%	29.2%	4.2x
15%	6,001	1,500	53.3%	25.0%	3.6x
20%	8,001	1,733	61.6%	21.7%	3.1x
30%	12,001	2,074	73.7%	17.3%	2.5x
40%	16,001	2,314	82.3%	14.5%	2.1x
50%	20,001	2,482	88.2%	12.4%	1.8x

The extra calls become less productive as more customers are flagged: from 15% to 20% about 12 of every 100 additional calls reach a bouncer, but from 40% to 50% only about 4 do.

Recommendation: flag the top 20% (raw score of at least 0.609). This reaches about 62% of bouncers and sits at the end of the steep part of the curve. A tiered approach can match cost to risk: call the top 10%, send an SMS to the next 20%, and take no action on the rest.

The final choice depends on three facts that must come from the bank: the monthly contact capacity of the collections team, the cost of one bounce, and the cost and success rate of one reminder.

9. Explainability

SHAP was used to explain the LightGBM model. SHAP shows how much each feature pushes an account's risk up or down from the average. BAL_TO_EMI was the most influential feature, followed by the balance and credit-movement features, which agrees with the EDA. For any single flagged account, a waterfall chart lists the reasons, for example a balance that covers less than one EMI and falling credit activity. This lets the collections team and risk reviewers see why an account was flagged.

10. Limitations and Next Steps

Limitations

1. **Synthetic data.** The data was generated from a simple rule plus noise, so these results demonstrate the pipeline, not real-world performance. The models are tied because the synthetic signal is simple. Real data usually has thresholds and interactions where tree-based models pull ahead.
2. **Target definition.** The exact observation window (for example, a bounce within the next 30 days) and the definition of a bounce must be confirmed with the bank.
3. **Leakage risk.** Every feature must be available before the due date. Any feature that already reflects the bounce would inflate the results.
4. **Scope.** Only standard accounts are modelled. SMA and NPA accounts were excluded.
5. **Cut-off drift.** A fixed score cut-off can drift as customer behaviour changes.

Next steps

1. Repeat the full comparison on the real bank data, including a time-based split (train on earlier months, test on later ones).
2. Confirm the cost figures and contact capacity, and set the threshold or tiers accordingly.
3. Monitor the monthly flag rate, the bounce rate in the flagged group and feature drift, and recalibrate when needed.
4. Build a monthly scoring job that loads the saved model and calibrator and writes results to an Oracle table for the collections team.

Appendix: Tools and Files

- **Tools:** Python (pandas, scikit-learn, LightGBM, XGBoost, SHAP, matplotlib, seaborn) in Jupyter Notebook, and Oracle SQL Developer for storage and SQL-based analysis.
- **Files:** emi_bounce_data.csv (synthetic data), generate_emi_bounce_data.py (data generator), oracle_emi_bounce.sql (table, checks and EDA queries), and the saved model file with its calibrator and feature list.

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